

Notes: Clerides, Lach, and Tybout (QJE, 1998)
Is Learning by Exporting Important? Micro-Dynamic Evidence from
Colombia, Mexico, and Morocco

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	3
3.1	Plant-level panels	3
3.2	Balanced panel and export-oriented industries	3
3.3	Plant types	3
3.4	Cost and productivity measures	4
3.5	Export spillovers	4
4	Models and empirical specifications	4
4.1	Exporting with sunk entry costs	4
4.2	Cost process	4
4.3	Dynamic export participation equation	5
4.4	FIML estimation	5
4.5	GMM robustness	5
4.6	Spillover equations	5
5	Identification and empirical design	5
5.1	Core identification challenge	5
5.2	Trajectory logic	6
5.3	Econometric learning test	6
5.4	Selection test	6
5.5	Spillover identification	6
5.6	Limitations	6
6	Section-by-section study guide	6
7	Figures and tables	8
7.1	Figure I: Gross Operating Profits from Exporting	8
7.2	Figures IIa and IIb: Simulated Average Cost Trajectories With and Without Learning	8
7.3	Tables I and II: Descriptive Sample Structure and Plant Types	9
7.4	Figures IIIa-IIIc: Path of Average Variable Cost	9
7.5	Figures IVa-IVc: Path of Average Labor Productivity	10
7.6	Tables IIIa and IIIb: FIML Estimates of Export Participation and Cost Equations	11

7.7	Tables IVa and IVb: GMM Estimates of the Cost Function	12
7.8	Table V and Tables VIa-VIb: Externalities and Spillovers	13
7.9	Tables VII and VIII: Appendix Data Tables	14
8	Detailed result map	15
9	Short summary	15
10	Conclusion	16
10.1	Core conclusion	16
10.2	Incremental contribution revisited	16
10.3	Policy implications	16
10.4	Limits and open questions	16
10.5	Final takeaway	16

1 Big picture

- **Question.** Do firms become more productive because they export, or do already productive firms select into exporting?
- **Policy motivation.** Export-promotion policies are often justified by learning-by-exporting: foreign buyers, foreign competition, and exposure to international best practice may raise firm efficiency. The paper asks whether this mechanism is visible in plant-level data from semi-industrialized countries.
- **Core distinction.** The positive cross-sectional exporter productivity premium can arise from two very different channels: *self-selection* of low-cost firms into export markets, or *learning* after firms enter export markets.
- **Empirical setting.** The paper uses plant-level panel data from Colombia, Mexico, and Morocco. It focuses on export-oriented industries and follows plants through entry into, exit from, and continued participation in foreign markets.
- **Descriptive strategy.** The authors plot productivity and cost trajectories for nonexporters, continuing exporters, export entrants, export quitters, and switchers. This visual comparison is guided by a dynamic model with sunk export-market entry costs.
- **Econometric strategy.** The paper estimates autoregressive cost functions together with dynamic export-participation equations. It tests whether lagged export participation shifts the cost process after controlling for lagged costs, capital, and selection into exporting.
- **Main result.** Relatively efficient firms become exporters, and firms that leave export markets tend to be less efficient. But in most industries, past exporting does not significantly reduce future costs. The exporter productivity premium is therefore mostly self-selection, not learning-by-exporting.
- **Spillover result.** There is some evidence that export activity in a region or industry may affect other firms' exporting probabilities or costs, especially in Colombia, but the evidence is mixed and not uniformly cost-reducing.
- **Economic takeaway.** Exporting status is not a sufficient proxy for productivity gains from exporting. The key empirical question is dynamic: do firms' cost processes improve after export-market entry? In this paper, usually they do not.

2 Incremental contribution of the paper

- **Separates learning from selection.** Earlier work showed that exporters are more productive than nonexporters. Clerides, Lach, and Tybout provide one of the classic micro-dynamic tests of whether exporting causes productivity improvement or whether productive firms self-select into exporting.
- **Dynamic plant-level framework.** The paper embeds exporting in a dynamic decision problem with sunk entry costs, export-market participation persistence, and autoregressive cost processes. This moves the literature beyond static exporter/nonexporter comparisons.
- **Cross-country plant evidence.** The paper compares three developing or semi-industrialized economies - Colombia, Mexico, and Morocco - rather than relying on one country or one industry. This makes the conclusion more general and highlights country-specific heterogeneity.
- **Trajectory-based visual diagnostic.** The simulated and actual trajectory figures are an important methodological contribution. They show what learning should look like in data: exporters should pull away from nonexporters after entering foreign markets.
- **Joint modeling of exporting and costs.** The FIML framework estimates export participation and cost processes simultaneously, allowing the authors to characterize selection and test for cost effects of lagged exporting.

- **GMM robustness.** The paper then relaxes the cost-function restrictions with GMM estimators, addressing concerns that misspecified autoregressive cost processes could falsely attribute cost dynamics to exporting.
- **Externality evidence.** The paper extends the analysis from own learning to regional and industry-level spillovers, helping distinguish private learning-by-exporting from possible external benefits of export activity.
- **Policy implication.** The paper weakens a simple rationale for export promotion: if export participation does not generally improve participating firms’ costs, then policy arguments must rely on selection, fixed-cost support, or spillovers rather than private learning alone.

3 Data and measurement

3.1 Plant-level panels

- Colombia: annual plant data for 1981–1991, covering plants with at least ten workers.
- Morocco: annual firm/plant data for 1984–1991, covering nearly all firms with at least ten workers.
- Mexico: annual data for larger firms over 1986–1990, with export information unavailable in the first two years.
- The authors use the term “plant” broadly because most plants in these data are single-plant firms.

Interpretation: The data are unusually rich for the period because they track individual production units, exporting, inputs, output, and costs over time. This makes dynamic export-entry and cost-process analysis possible.

3.2 Balanced panel and export-oriented industries

- The estimation sample includes plants reporting the relevant variables throughout the sample period.
- The main analysis focuses on export-oriented industries: industries exporting at least 10 percent of output and with at least twenty exporting plants, with some exceptions for important export sectors.
- The balanced-panel restriction excludes many small entrants and exiters, but those plants account for small shares of output.

Interpretation: The balanced panel improves dynamic estimation but raises selection concerns. The paper directly discusses this tradeoff and argues that excluded plants are mostly small, so aggregate influence is limited.

3.3 Plant types

The descriptive analysis divides plants into five categories:

- **Nonexporters:** never exported during the sample.
- **Exporters:** exported throughout the sample.
- **Entrants:** start as nonexporters and begin exporting.
- **Quitters:** start as exporters and stop exporting.
- **Switchers:** change export status more than once.

Interpretation: This classification makes the learning-by-exporting test concrete. If learning is important, entrants should show improving cost/productivity trajectories after their transition year.

3.4 Cost and productivity measures

- Average variable cost (AVC) is real labor plus real intermediate input costs divided by real output.
- Labor productivity (LAB) is real output divided by number of workers.
- Both measures are purged of industry-year effects and observable plant characteristics such as age, capital, and size where available.
- Residualized cost/productivity trajectories are then compared across plant types and transition years.

Interpretation: AVC is the main cost measure because the formal model is about cost processes. Labor productivity is used as a complementary performance measure because AVC may reflect input-price or skill-mix differences.

3.5 Export spillovers

- Industry-level spillovers measure export activity by other firms in the same industry.
- Region-level spillovers measure export activity by other firms in the same geographic region.
- The paper uses both counts of exporters and export volumes as alternative spillover measures.

Interpretation: These variables test whether exporters create external effects that benefit other firms, either by transmitting knowledge or by improving input, transport, and export-service markets.

4 Models and empirical specifications

4.1 Exporting with sunk entry costs

A firm participates in export markets when expected export profits exceed domestic opportunity costs and the fixed or sunk cost of export-market entry. The dynamic participation rule can be summarized as:

$\text{Export}_{it} = 1$ if expected value of exporting exceeds expected value of not exporting net of start-up costs.

Interpretation: Sunk costs generate persistence. A plant may continue exporting even when current export profits are weak because leaving would force it to pay reentry costs later.

4.2 Cost process

The paper's core learning test is based on an autoregressive cost equation of the form:

$$\ln AVC_{it} = \alpha_i + \rho_1 \ln AVC_{i,t-1} + \rho_2 \ln AVC_{i,t-2} + \beta_1 y_{i,t-1} + \beta_2 y_{i,t-2} + \beta_3 y_{i,t-3} + X'_{it} \gamma + u_{it}.$$

- y_{it} indicates export-market participation.
- X_{it} includes capital, age, output, industry-year controls, and other covariates depending on specification.
- Learning-by-exporting predicts negative coefficients on lagged export participation in the cost equation.

Interpretation: The empirical question is whether export history shifts the cost process after controlling for past costs. This is a Granger-causality-style test for learning.

4.3 Dynamic export participation equation

The export participation equation includes lagged cost, capital, real exchange rates, age, and previous export experience:

$$\Pr(y_{it} = 1) = F(\text{capital}_{i,t-1}, \ln AVC_{i,t-1}, \ln AVC_{i,t-2}, y_{i,t-1}, y_{i,t-2}, y_{i,t-3}, e_t, A_{it}).$$

Interpretation: The export equation tests self-selection. Low-cost, large-capital, and experienced plants should be more likely to export if foreign-market entry involves fixed or sunk costs.

4.4 FIML estimation

The paper jointly estimates the export participation equation and cost equation using full information maximum likelihood for Colombia and Morocco.

(participation equation, cost equation) estimated jointly with correlated disturbances.

Interpretation: Joint estimation allows selection and cost shocks to be related. This is important because plant cost shocks affect export entry and export entry may be correlated with unobserved cost heterogeneity.

4.5 GMM robustness

The paper then estimates more flexible cost functions using GMM. This allows richer dynamics, additional lags, and broader control sets:

$$E[Z'_{it}u_{it}] = 0,$$

where Z_{it} includes instruments based on lagged variables.

Interpretation: The GMM results test whether the no-learning conclusion is an artifact of a restrictive FIML cost specification.

4.6 Spillover equations

The spillover specifications augment the cost or participation equations with regional or industry export activity:

$$\ln AVC_{it} = \dots + \delta_R \text{ExportActivity}_{r,t-1} + \delta_J \text{ExportActivity}_{j,t-1} + u_{it}.$$

Interpretation: If exporters improve regional suppliers, transport services, or knowledge pools, then non-exporters in high-export regions or industries may experience lower costs or higher export-entry probabilities.

5 Identification and empirical design

5.1 Core identification challenge

Exporters are not randomly selected. They are likely to be larger, lower-cost, more capital intensive, and more likely to face favorable foreign demand shocks. A raw exporter productivity premium therefore cannot identify learning-by-exporting.

5.2 Trajectory logic

- If learning matters, entrants should become more efficient after export entry.
- If selection matters, entrants should already be efficient before export entry, and their post-entry trajectories need not improve.
- If sunk costs matter, past export status should strongly predict current export status.

Interpretation: This is why the paper emphasizes transition-year plots. Learning is a dynamic prediction, not a cross-sectional level prediction.

5.3 Econometric learning test

The formal test asks whether past export participation enters the cost function after controlling for lagged costs and state variables. The key null is:

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0.$$

Learning-by-exporting predicts negative coefficients because exporting history should lower future costs.

5.4 Selection test

The participation equation tests whether low past costs and export experience predict export status. Persistent export participation, large capital stocks, and low costs are evidence of selection into exporting.

5.5 Spillover identification

Spillover estimates use time variation in regional or industry export activity. The challenge is that export activity may respond to regional factor-cost shocks, so negative spillover coefficients can reflect either true externalities or correlated regional cost declines.

5.6 Limitations

- Balanced panels exclude many small and transitory firms.
- AVC and labor productivity are imperfect measures of total factor productivity.
- Export demand shocks may be firm-specific and difficult to separate from productivity shocks.
- Spillovers are hard to identify because regional export growth can coincide with regional cost changes.
- The results apply to manufacturing industries and countries available in the data, not necessarily all export sectors.

6 Section-by-section study guide

- **Introduction.** Motivates the learning-by-exporting hypothesis and separates learning from self-selection.
- **Theory and simulations.** Develops the dynamic export decision with sunk costs and shows what cost trajectories should look like with and without learning.
- **Data and descriptive trajectories.** Defines plant types and plots average cost and labor productivity around export-market entry/exit.

- **Econometric tests.** Estimates export-participation and cost equations with FIML and GMM.
- **Spillovers.** Tests whether regional or industry export activity affects export participation or costs.
- **Conclusion.** Argues that self-selection explains most exporter productivity premia; private learning-by-exporting is weak, while spillovers are possible but mixed.

7 Figures and tables

7.1 Figure I: Gross Operating Profits from Exporting

Read:

- The figure plots domestic and foreign sales against marginal cost.
- Exporting becomes profitable below a cost threshold.
- This illustrates self-selection: low-cost plants have stronger incentives to export.

Detailed interpretation: The figure is the theoretical foundation for the self-selection channel. Even without learning, low-cost firms will enter foreign markets because the marginal returns to exporting are higher for them. Therefore, a positive exporter productivity premium is not evidence of learning.

Economic message: The first lesson is selection: efficient plants export because exporting is more profitable for them, not necessarily because exporting makes them efficient.

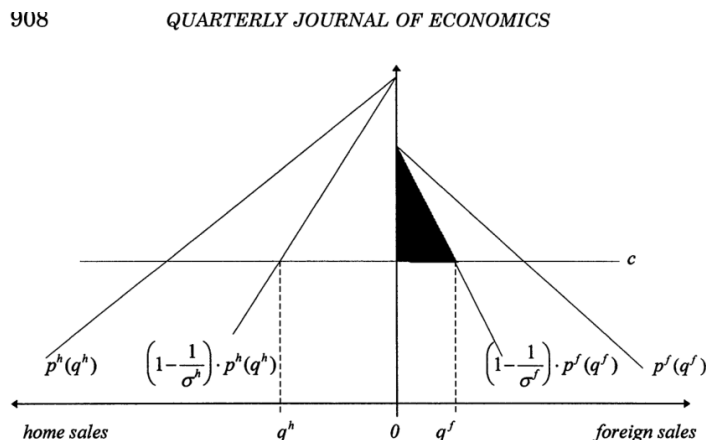


FIGURE I
Gross Operating Profits from Exporting

Figure I: Gross Operating Profits from Exporting

7.2 Figures IIa and IIb: Simulated Average Cost Trajectories With and Without Learning

Read:

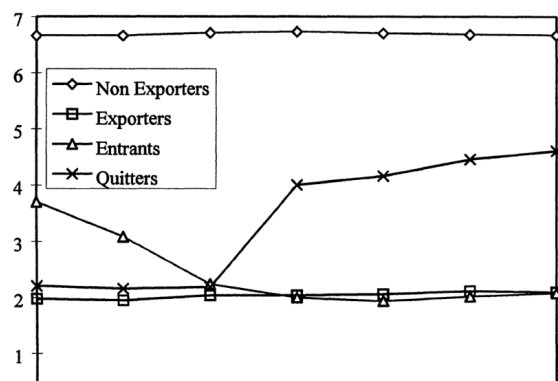
- Figure IIa shows simulated cost paths without learning.
- Figure IIb shows simulated cost paths with learning.
- Without learning, entrants are already relatively low cost before entry and do not continue to pull away afterward.
- With learning, entrants continue to reduce costs after entering export markets.

Detailed interpretation: These figures provide the paper's diagnostic template. They show what the data should look like if learning is present. A learning effect is not simply that entrants are efficient; it is that entrants improve after entry relative to the industry norm.

Economic message: The key empirical signature of learning-by-exporting is post-entry improvement, not pre-entry superiority.

912

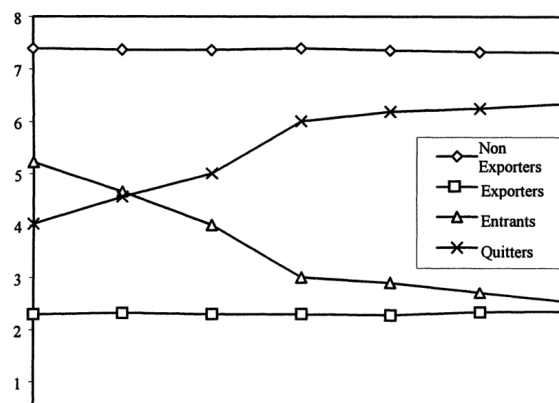
QUARTERLY JOURNAL OF ECONOMICS



(a) Figure IIa: Simulated cost trajectories without learning

IS LEARNING BY EXPORTING IMPORTANT?

913



(b) Figure IIb: Simulated cost trajectories with learning

7.3 Tables I and II: Descriptive Sample Structure and Plant Types

Read:

- Table I reports average entry rates, exit rates, plant counts, and export intensity for export-oriented industries.
- Mexico and Morocco have higher entry rates than Colombia in the analysis sample.
- Morocco has the highest average export intensity.
- Table II defines the five plant categories used in the trajectory analysis.

Detailed interpretation: These tables set up the descriptive analysis. The authors need enough export-market transitions to compare entrants and quitters; Table I shows those transitions exist. Table II makes the transition definitions transparent.

Economic message: The paper's empirical strategy depends on plant transitions. Tables I and II show the variation and taxonomy needed for that strategy.

IS LEARNING BY EXPORTING IMPORTANT?

915

TABLE I
ENTRY, EXIT, NUMBER OF PLANTS, AND EXPORT INTENSITY BY COUNTRY
(EXPORT-ORIENTED INDUSTRIES)

	Average annual entry rate	Average annual exit rate	Average number of plants	Average export intensity
Colombia 1981–1991	.027	.017	1354	.095
Morocco 1984–1991	.049	.037	938	.360
Mexico 1986–1990	.048	.015	1327	.230

(a) Table I: Entry, exit, number of plants, and export intensity

916

QUARTERLY JOURNAL OF ECONOMICS

TABLE II
TYPES OF FIRMS

<i>Nonexporters:</i>	firms that never exported during the sample period.
<i>Exporters:</i>	firms that always exported during the sample period.
<i>Entrants:</i>	firms that began the period as nonexporters, but began exporting during the sample period and never stopped.
<i>Quitters:</i>	firms that began the period as exporters, but ceased during the sample period and never resumed exporting.

(b) Table II: Types of firms

7.4 Figures IIIa-IIIc: Path of Average Variable Cost

Read:

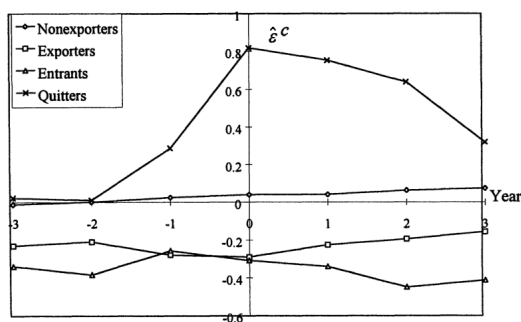
- Figures IIIa-IIIc plot residual average variable cost paths for Colombia, Mexico, and Morocco.
- Export entrants generally have relatively low costs before export entry.
- Entrants do not show strong and systematic post-entry cost reductions.
- Quitters tend to have worsening cost performance before leaving export markets.

Detailed interpretation: The figures look more like the no-learning simulation than the learning simulation. Entrants are selected from the low-cost part of the plant distribution before entry. After entry, there is little evidence that their cost trajectories improve further.

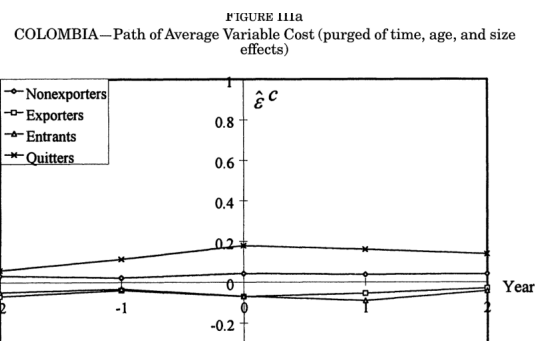
Economic message: The cost trajectories support self-selection much more than private learning-by-exporting.

918

QUARTERLY JOURNAL OF ECONOMICS



(a) Figure IIIa: Colombia - average variable cost



(b) Figure IIIb: Mexico - average variable cost

IS LEARNING BY EXPORTING IMPORTANT?

919

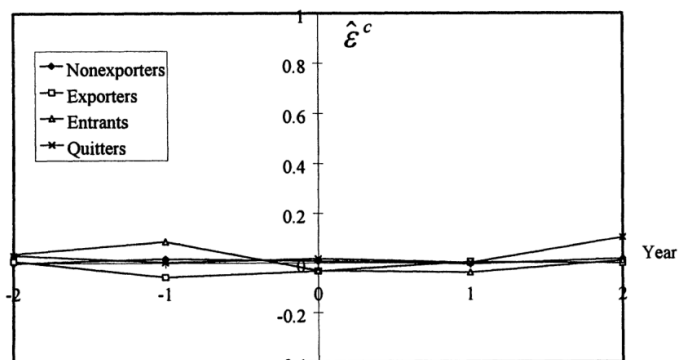


Figure IIIc: Morocco - average variable cost

7.5 Figures IVa-IVc: Path of Average Labor Productivity

Read:

- Figures IVa-IVc repeat the trajectory analysis using labor productivity.
- Colombian entrants show some improvement in labor productivity around export entry.

- Mexican entrants do not show a learning pattern.
- Moroccan labor productivity patterns differ from average cost patterns, suggesting input-mix and wage-cost interactions.

Detailed interpretation: Labor productivity provides the best evidence for learning in Colombia, but the pattern is not general across countries. The paper interprets this cautiously because labor productivity can reflect skill composition and capital intensity, not only technical learning.

Economic message: Some country-industry cases look consistent with learning, but the broad cross-country evidence is weak.

920

QUARTERLY JOURNAL OF ECONOMICS

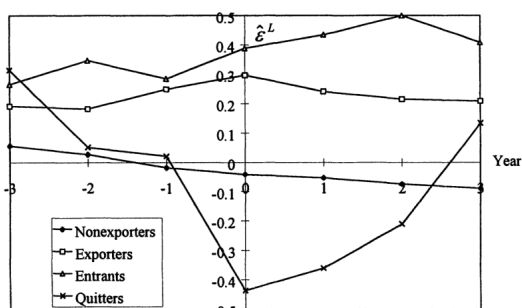
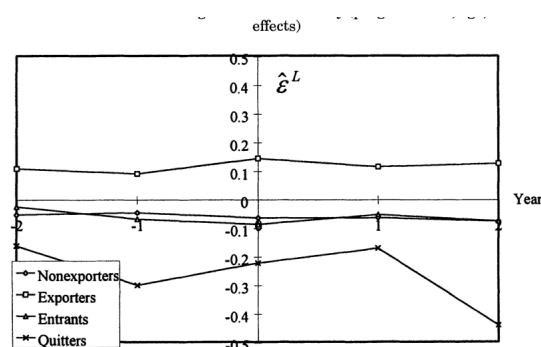


FIGURE IVa
COLOMBIA—Path of Average Labor Productivity (purged of time, age, and size)

(a) Figure IVa: Colombia - labor productivity



(b) Figure IVb: Mexico - labor productivity

IS LEARNING BY EXPORTING IMPORTANT?

921

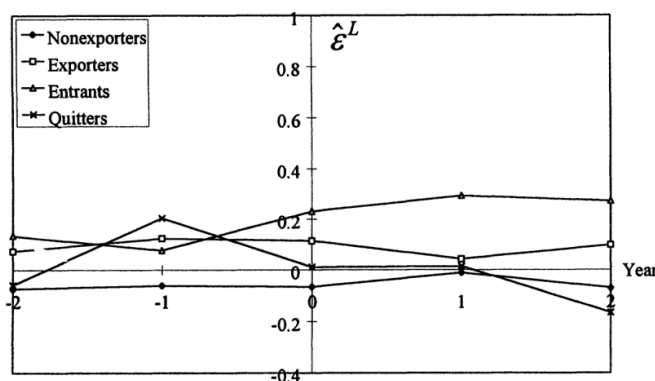


Figure IVc: Morocco - labor productivity

7.6 Tables IIIa and IIIb: FIML Estimates of Export Participation and Cost Equations

Read:

- Tables IIIa and IIIb jointly estimate export participation and average variable cost equations for Colombia and Morocco.

- Lagged export status strongly predicts current export participation, consistent with sunk export-market costs.
- Low costs and larger capital stocks tend to predict exporting.
- Lagged export participation generally does not reduce future costs; where significant, some coefficients have the wrong sign.

Detailed interpretation: These are the paper's core formal tests. The participation equation supports self-selection and persistence. The cost equation mostly rejects private learning because export history does not lower costs after controlling for lagged costs and plant state variables.

Economic message: The FIML evidence says: efficient firms choose exporting, but exporting usually does not make firms more efficient.

928

QUARTERLY JOURNAL OF ECONOMICS

TABLE IIIa
FIML ESTIMATES OF EQUATIONS (10) AND (11)
COLOMBIA 1983-1991

	Chemicals	Textiles	Apparel
Participation equation			
Intercept	-15.06 (5.87)*	-4.34 (1.33)*	-5.20 (1.58)*
$\ln(e_t)$	.462 (2.27)	3.93 (1.14)*	3.60 (0.94)*
$\ln(K_{it-1})$	3.73 (0.72)*	1.71 (0.34)*	2.03 (0.37)*
$\ln(A_{it})$	5.94 (3.87)	-0.12 (0.85)	0.64 (1.07)
$\ln(A_{it})^2$	-0.88 (0.58)	0.09 (0.14)	-0.01 (0.18)
B_{it}	-0.00 (0.28)	0.23 (0.15)	0.28 (0.14)*
$\ln(AVC_{it-1})$	-0.14 (0.28)	-0.18 (0.18)	-0.05 (0.11)
$\ln(AVC_{it-2})$	0.06 (0.29)	-0.16 (0.18)	-0.10 (0.13)
$\tilde{y}_{it-1}$	1.86 (0.41)*	2.04 (0.27)*	1.03 (0.25)*
$\tilde{y}_{it-2}$	0.18 (0.40)	0.25 (0.29)	-0.10 (0.25)
$\tilde{y}_{it-3}$	0.26 (0.40)	-0.57 (0.46)	-0.20 (0.25)
Cost function			
Intercept	-0.16 (0.13)	-0.33 (0.06)*	-0.36 (0.06)*
$\ln(K_{it})$	-0.17 (0.11)	0.05 (0.04)	0.09 (0.06)
$\ln(AVC_{it-1})$	0.36 (0.05)*	0.65 (0.02)*	0.74 (0.02)*
$\ln(AVC_{it-1})$	0.45 (0.05)*	0.16 (0.03)*	0.12 (0.03)*
y_{it-1}	0.09 (0.05)	0.01 (0.03)	0.08 (0.03)*
y_{it-2}	0.22 (0.09)*	0.04 (0.06)	0.01 (0.08)
y_{it-3}	0.11 (0.10)	0.06 (0.07)	0.01 (0.08)
Variance-covariance matrix for disturbances			
$\text{var}(\alpha_1)$	0.298	0.167	0.574
$\text{var}(\alpha_2)$	0.005	0.0001	0.001
$\text{corr}(\alpha_1, \alpha_2)$	-0.106	-0.147	-0.082
$\text{var}(\epsilon_1)$	0.702	0.823	0.426
$\text{var}(\epsilon_2)$	0.995	0.106	0.163
$\text{corr}(\epsilon_1, \epsilon_2)$	0.180	-0.110	-0.026
No. observations	567	1,854	2,547
Log-likelihood	-309.61	-938.45	-1,679.24

(a) Table IIIa: FIML estimates, Colombia

IS LEARNING BY EXPORTING IMPORTANT?

929

TABLE IIIb
FIML ESTIMATES OF EQUATIONS (10) AND (11)
MOROCCO 1984-1990

	Chemicals	Food	Textiles
Participation equation			
Intercept	17.66 (18.18)	16.35 (15.04)	16.49 (10.05)
$\ln(e_t)$	5.69 (3.90)	3.58 (3.22)	3.50 (2.13)
$\ln(K_{it-1})$	2.64 (0.67)*	1.03 (0.46)*	2.45 (0.43)*
$\ln(A_{it})$	2.45 (3.23)	-1.56 (1.73)	-2.09 (1.47)
$\ln(A_{it})^2$	-0.39 (0.48)	0.28 (0.26)	0.35 (0.24)
B_{it}	0.81 (0.70)	0.26 (0.15)	0.07 (0.16)
$\ln(AVC_{it-1})$	-1.16 (0.60)	-0.18 (0.42)	0.06 (0.18)
$\ln(AVC_{it-2})$	-1.05 (0.89)	0.92 (0.47)	-0.24 (0.30)
$\tilde{y}_{it-1}$	1.14 (0.45)*	1.25 (0.54)*	0.91 (0.36)*
$\tilde{y}_{it-2}$	0.09 (0.05)	1.25 (0.51)*	0.50 (0.28)
$\tilde{y}_{it-3}$	0.28 (0.35)	0.67 (0.47)	0.02 (0.28)
Cost function			
Intercept	0.05 (0.03)	-0.07 (0.03)*	-0.12 (0.03)*
$\ln(K_{it})$	-0.06 (0.04)	-0.16 (0.04)*	-0.07 (0.04)
$\ln(AVC_{it-1})$	0.39 (0.05)*	0.20 (0.05)*	0.15 (0.04)*
$\ln(AVC_{it-1})$	0.40 (0.07)*	0.25 (0.05)*	0.07 (0.05)
y_{it-1}	-0.02 (0.03)	0.12 (0.02)*	-0.02 (0.02)
y_{it-2}	0.03 (0.05)	0.02 (0.04)	0.00 (0.05)
y_{it-3}	0.02 (0.04)	0.01 (0.10)	-0.12 (0.06)
$\text{var}(\alpha_1)$	0.546	0.724	0.677
$\text{var}(\alpha_2)$	0.0001	0.003	0.001
$\text{corr}(\alpha_1, \alpha_2)$	0.046	-0.559	-0.590
$\text{var}(\epsilon_1)$	0.454	0.276	0.323
$\text{var}(\epsilon_2)$	0.024	0.022	0.057
$\text{corr}(\epsilon_1, \epsilon_2)$	-0.019	-0.040	-0.057
No. observations	637	1,169	1,722
Log-likelihood	69.13	117.79	-517.87

Standard errors are in parentheses. An asterisk indicates that the estimate is significant at a 95 percent significance level.

(b) Table IIIb: FIML estimates, Morocco

7.7 Tables IVa and IVb: GMM Estimates of the Cost Function

Read:

- Tables IVa and IVb relax the cost-function restrictions used in FIML.
- They include more flexible dynamics and richer controls.
- Sargan and serial-correlation tests are reported for specification checking.
- The learning-by-exporting coefficients remain generally insignificant or inconsistent with cost reduction.

Detailed interpretation: These tables are important robustness tests. They address the concern that FIML could miss higher-order cost dynamics and falsely reject learning. The results still do not support

broad private learning-by-exporting.

Economic message: The no-learning conclusion survives more flexible dynamic panel estimation.

TABLE IVa
GMM ESTIMATES OF COST FUNCTION (11'), COLOMBIA, 1986-1991

	Apparel	Other chemicals	Textiles	Paper	Industrial chemicals	Nonbeverage food processing	Beverages
Exporting experience measured with $y_{j,t-1}, y_{j,t-2}, y_{j,t-3}$							
Coefficient on:							
$y_{j,t-1}$	0.021 (0.041)	-0.015 (0.027)	0.018 (0.027)	-0.041 (0.035)	0.012 (0.027)	-0.030 (0.030)	-0.083 (0.033)
$y_{j,t-2}$	-0.010 (0.040)	-0.013 (0.026)	0.004 (0.025)	-0.008 (0.020)	0.002 (0.016)	0.040 (0.023)	0.101 (0.044)
$y_{j,t-3}$	0.081 (0.043)	0.048 (0.033)	0.008 (0.025)	0.050 (0.030)	0.029 (0.021)	0.017 (0.025)	0.014 (0.016)
Wald test, joint signif., $\chi^2(3)$	9.07*	2.24	2.29	3.95	4.49	7.25	7.77
Sargan test, instr., $\chi^2(3)$	34.11*	34.18*	30.62	27.02	31.04	26.53	29.23
1st order autocorr., $N(0,1)$	-0.727	0.850	-1.737	-0.95	-0.100	-0.487	-0.138
2nd order autocorr., $N(0,1)$	-0.830	0.882	-0.936	-1.44	1.002	-1.077	-1.751
Exporting experience measured with $\ln(q'_{j,t-1} + 1) \ln(q'_{j,t-2} + 1) \ln(q'_{j,t-3} + 1)$							
Coefficient on:							
$\ln(q'_{j,t-1} + 1)$	0.002 (0.005)	-0.003 (0.002)	0.003 (0.003)	-0.006 (0.004)	0.002 (0.003)	-0.004 (0.003)	-0.006 (0.005)
$\ln(q'_{j,t-2} + 1)$	-0.004 (0.006)	-0.001 (0.003)	0.002 (0.003)	-0.001 (0.002)	-0.001 (0.002)	0.004 (0.003)	0.006 (0.005)
$\ln(q'_{j,t-3} + 1)$	0.010 (0.006)	0.006 (0.004)	-0.002 (0.003)	0.008 (0.004)	0.001 (0.002)	0.002 (0.003)	0.001 (0.002)*
Wald test, joint signif., $\chi^2(3)$	10.41*	3.21	3.59	4.32	3.40	7.98*	6.81
Sargan test, instr., $\chi^2(3)$	23.94	24.35	30.87	26.33	30.31	26.58	29.442
1st order autocorr., $N(0,1)$	-0.711	0.984	-1.749	-0.994	-0.100	-0.480	-0.141
2nd order autocorr., $N(0,1)$	-0.809	0.951	-0.948	-1.437	1.007	-1.058	-1.744
No. plants	283	169	206	69	63	301	93
No. observations	1698	1014	1236	414	378	1806	558

Standard errors are in parentheses; an asterisk indicates that the estimate is significant at a 95 percent significance level.
In addition to measures of exporting experience, the regressions include the following variables: three lags of cost, three lags of capital stock, three lags of output, current output, business type, age, age squared, and annual time dummies. All are included in the instrument set except current output.
The Wald test is for the joint null hypothesis that the three variables measuring previous exporting experience have no coefficients.
The Sargan test, which is based on overidentifying restrictions, is for the null hypothesis that the instrument set is orthogonal to the disturbance.
The autocorrelation tests are for first- and second-order serial correlation in the (unfixed-effects) residuals.

(a) Table IVa: GMM estimates, Colombia

TABLE IVb
GMM ESTIMATES OF COST FUNCTION (11'), MOROCCO, 1988-1990

	Food	Textiles	Apparel	Leather	Metalsworking	Chemicals
Exporting experience measured with $y_{j,t-1}, y_{j,t-2}, y_{j,t-3}$						
Coefficient on:						
$y_{j,t-1}$	0.108 (0.068)	-0.020 (0.023)	0.033 (0.033)	-0.031 (0.040)	-0.008 (0.046)	-0.010 (0.018)
$y_{j,t-2}$	0.017 (0.041)	0.037 (0.022)	-0.018 (0.029)	-0.000 (0.030)*	-0.022 (0.041)	-0.009 (0.016)
$y_{j,t-3}$	-0.051 (0.051)	-0.042 (0.022)	-0.049 (0.024)*	0.042 (0.035)	-0.027 (0.066)	-0.007 (0.018)
Wald test, joint signif., $\chi^2(3)$	29.92*	4.35	6.66	9.49*	4.51	9.95
Sargan test, instr., $\chi^2(5)$	7.40	10.84	7.50	6.63	4.27	6.65
1st order autocorr., $N(0,1)$	-0.584	-0.754	-0.600	-0.022	-1.711	-0.662
2nd order autocorr., $N(0,1)$	0.040	0.453	0.771	-0.000	-1.132	0.395
Exporting experience measured with $\ln(q'_{j,t-1} + 1) \ln(q'_{j,t-2} + 1) \ln(q'_{j,t-3} + 1)$						
Coefficient on:						
$\ln(q'_{j,t-1} + 1)$	0.013 (0.009)	-0.003 (0.007)	0.010 (0.008)	0.007 (0.010)	-0.001 (0.008)	-0.002 (0.003)
$\ln(q'_{j,t-2} + 1)$	0.002 (0.004)	0.001 (0.004)	-0.008 (0.005)	-0.009 (0.005)	-0.002 (0.015)	-0.001 (0.003)
$\ln(q'_{j,t-3} + 1)$	-0.007 (0.007)	-0.002 (0.003)	-0.008 (0.004)	0.005 (0.006)	-0.004 (0.011)	-0.000 (0.003)
Wald test, joint signif., $\chi^2(3)$	27.65*	1.89	9.11*	16.41*	4.05	4.90
Sargan test, instr., $\chi^2(5)$	7.09	10.47	7.74	6.09	4.28	5.44
1st order autocorr., $N(0,1)$	-0.681	-0.840	-0.705	0.002	-1.596	-0.562
2nd order autocorr., $N(0,1)$	-0.940	0.599	0.780	0.096	-1.194	0.352
No. plants	167	246	157	105	116	91
No. observations	501	738	471	315	548	273

Standard errors are in parentheses; an asterisk indicates that the estimate is significant at a 95 percent significance level.
In addition to measures of exporting experience, the regressions include the following variables: three lags of cost, three lags of capital stock, three lags of output, current output, business type, age, age squared, and annual time dummies. All are included in the instrument set except current output.
The Wald test is for the joint null hypothesis that the three variables measuring previous exporting experience have no coefficients.
The Sargan test, which is based on overidentifying restrictions, is for the null hypothesis that the instrument set is orthogonal to the disturbance.
The autocorrelation tests are for first- and second-order serial correlation in the (unfixed-effects) residuals.

(b) Table IVb: GMM estimates, Morocco

7.8 Table V and Tables VIa-VIb: Externalities and Spillovers

Read:

- Table V estimates spillover effects for Colombia using FIML.
- Tables VIa and VIb use GMM specifications for Colombia and Morocco.
- Regional export activity is sometimes associated with lower costs, especially in Colombia.
- Industrywide export activity can be cost-reducing in some cases and cost-increasing in others.

Detailed interpretation: The spillover evidence is more mixed than the self-selection evidence. The strongest interpretation is that export activity can sometimes affect the regional production environment, but estimated effects differ by country and industry. Regional spillovers may capture genuine externalities or correlated regional shocks.

Economic message: Export promotion may still have a spillover rationale, but the evidence is less clean than simple learning-by-exporting stories suggest.

IS LEARNING BY EXPORTING IMPORTANT?

943

TABLE VII
INDUSTRIES AND NUMBERS OF PLANTS IN THE DATABASE, BY COUNTRY

Colombia	Morocco	Mexico
Coffee processing (127)	Food processing (937)	Food processing (376)
Other food processing (965)	Textiles (564)	Textiles (185)
Textiles (457)	Apparel (671)	Chemicals (395)
Apparel (962)	Leather products (292)	Glass (21)
Leather products (94)	Transport equipment (103)	Metal products (166)
Paper (353)	Metalworking (354)	Nonelectrical machinery (151)
Industrial chemicals (113)	Chemicals, nonphosphate (285)	Electric machinery (142)
Other chemicals (284)	Phosphate products (324)	Transport equipment (149)
		Electronic components

(a) Table VII: Industries and plant counts

TABLE VIII
COMPONENTS OF TOTAL VARIABLE COST, BY COUNTRY

Colombia	Morocco	Mexico
<u>Employee compensation:</u> • salaries • benefits	Cost of personnel	<u>Total labor remuneration:</u> • salaries • fringe benefits • social security contributions
<u>Cost of intermediate inputs:</u> • value of raw materials consumed • goods purchased for sale without transformation • purchases of replacement parts of less than one year duration • energy consumed • other fuels and lubricants	Materials and other non-capital expenditures	<u>Cost of materials:</u> • primary/raw materials consumed • packaging materials • fuels • electricity • spare parts
<u>Cost of services and transport:</u> • payments for industrial work by other establishments • payments to third parties		<u>Cost of services and transport:</u> • cost of subcontracted assembly work • cost of shipping • sales commissions

(b) Table VIII: Variable cost components

8 Detailed result map

- **Selection result.** Low-cost and larger plants are more likely to export; lagged exporting strongly predicts current exporting.
- **Private learning result.** Past export participation generally does not lower future costs once lagged costs and state variables are controlled.
- **Trajectory evidence.** Entrants usually look efficient before entering export markets and do not systematically improve afterward.
- **Labor productivity evidence.** Colombia shows some post-entry labor productivity gains, but this does not generalize across countries.
- **Externality evidence.** Regional or industry export activity sometimes predicts lower costs or entry probabilities, but effects are mixed.
- **Policy implication.** Export promotion should not be justified simply by assuming private learning; selection and spillovers must be separated empirically.

9 Short summary

- The paper studies whether firms learn by exporting using plant-level data from Colombia, Mexico, and Morocco.
- Exporters are more efficient than nonexporters.
- But efficient plants often become exporters before any possible learning effect occurs.
- Export entrants do not generally show continuing post-entry cost reductions.
- Formal FIML and GMM estimates usually reject a strong private learning-by-exporting effect.
- Export-market participation is persistent, consistent with sunk costs.
- There is mixed evidence of externalities from regional and industry export activity.

- The main conclusion is that exporter productivity premia mostly reflect self-selection rather than learning-by-exporting.

10 Conclusion

10.1 Core conclusion

Clerides, Lach, and Tybout find that exporters are more efficient, but the causal interpretation is mostly selection rather than learning. Relatively efficient plants become exporters; exporting itself does not usually shift firms onto a better cost trajectory.

10.2 Incremental contribution revisited

The paper's incremental contribution is to turn the learning-by-exporting debate into a dynamic micro-econometric question. Instead of comparing exporters with nonexporters, it asks whether export-market entry changes the stochastic process for costs and productivity. This methodology influenced later trade research on exporter premia, firm selection, and productivity dynamics.

10.3 Policy implications

If the exporter productivity premium is mostly selection, then export promotion aimed at making incumbent exporters more efficient may have limited private returns. Policies that reduce fixed costs of entry may expand exporting, but their welfare case depends on whether social spillovers or market failures are present.

10.4 Limits and open questions

The paper uses balanced panels and imperfect productivity proxies. It cannot fully rule out learning in specific industries, especially Moroccan apparel and leather, or broader spillovers through suppliers and services. It also studies countries and time periods for which data were available, so later global value-chain settings may differ.

10.5 Final takeaway

The central lesson is methodological and substantive: exporter premia are not learning effects unless firms improve after entering export markets. In this paper, the strongest evidence is that productive firms self-select into exporting.